Data Governance Framework

Michelin Sri Lanka Manufacturing Plant

Project by Group 14 of IS2209 Data Management and Governance Course at UCSC

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1. Organization Overview and Data Landscape

1.1 Organization Overview

The selected organization is the **Michelin Sri Lanka manufacturing plant**, which operates as part of the Michelin Group. The plant produces highly specialized industrial products and operates in a data-intensive manufacturing environment.

Key characteristics of the organization:

- Manufactures solid tyres, tracks, and industrial wheels
- Operates a make-to-order production model
- Handles more than 800 tyre product variations
- Requires strict quality control and traceability

1.2 Manufacturing Process and Data Life Cycle

The data life cycle at the plant closely aligns with the manufacturing process, starting from customer order intake and ending with finished goods dispatch. Data is continuously generated, processed, stored, and analysed at multiple stages of production.

End-to-end manufacturing data flow:

- Customer requirements collected by sales agents
- Orders and 3-month production plans maintained in ERP (SAP)
- Semi-finished materials ordered through ERP
- Production instructions passed to SCADA and MES
- Machines controlled via PLCs and HMIs
- Real-time sensor data generated during production
- Quality inspection data captured
- Finished goods dispatched and inventory updated

1.3 Data Landscape and Data Sources

The organization operates a complex data landscape consisting of both IT and OT systems. These systems must work together seamlessly to support production, quality, and performance monitoring.

Primary data sources include:

- ERP system (SAP) for orders, inventory, and dispatch
- SCADA and MES systems for production execution and recipes
- PLCs and sensors for real-time machine data
- HMIs for manual inputs and overrides
- Quality inspection and traceability systems
- Central database cluster and data lake
- Business intelligence tools such as Power BI

1.4 Current Data Governance and Management Practices

The organization has already implemented basic governance controls, especially for data that is critical to product quality and intellectual property protection. However, these practices are mostly system-specific rather than enterprise-wide.

Existing governance and management practices:

- Role-based access to product recipes
- Data owners responsible for recipe modifications
- Version control and audit logs for recipe changes

- Serial-number-based product traceability
- KPI monitoring for plant and product performance

1.5 Key Data Management Challenges

Despite existing controls, several challenges affect data consistency, usability, and scalability. These challenges highlight the need for a structured and integrated Data Governance and Management Framework.

Major challenges identified:

- Lack of standardized naming conventions across machines and factories
- Different data structures and acronyms for similar parameters
- High dependency on complex data dictionaries for analysis
- Integration challenges with legacy systems
- High cybersecurity investment requirements
- Limited availability of skilled local support personnel

2. Importance of Data Governance and Management for Michelin Sri Lanka

2.1 Need for Strong Data Governance

Michelin Sri Lanka operates in a highly complex manufacturing environment where production quality, consistency, and efficiency are heavily dependent on data. Without a structured Data Governance and Management approach, the organization risks data inconsistency, quality issues, and operational inefficiencies.

Key reasons data governance is essential:

- Large number of product variants (800+ tyres)
- Heavy reliance on machine and sensor data
- Multiple interconnected IT and OT systems
- Increasing data volumes and complexity

2.2 Protection of Intellectual Property and Critical Data

Product recipes, formulations, and R&D data represent the organization's most valuable intellectual assets. Any unauthorized access, modification, or leakage of this data can result in significant competitive and financial risks.

Why governance is critical for IP protection:

- Recipes directly affect product quality and safety
- Unauthorized changes can lead to defects or recalls
- Data classification enables stronger controls for super-critical data
- Encryption, access control, and audit logs reduce misuse

2.3 Ensuring Product Quality and Traceability

Michelin Sri Lanka maintains strict quality standards and relies on end-to-end traceability to investigate defects and customer complaints. Effective data management ensures that quality data is accurate, complete, and easily retrievable.

Governance supports quality by enabling:

- Serial-number-level traceability for each tyre
- Accurate linking of machine parameters to products
- Faster root-cause analysis for quality issues
- Improved compliance with internal quality standards

2.4 Improving Operational Efficiency and Performance Monitoring

Plant and product performance are continuously monitored using KPIs derived from multiple data sources. Without governance, inconsistent data definitions and calculations can lead to unreliable insights.

Benefits of governed data for operations:

- Consistent KPI definitions across departments
- Reliable plant and product performance metrics
- Reduced manual data correction and reconciliation
- Better decision-making for production planning

2.5 Supporting Advanced Analytics and Cloud Readiness

The organization plans to move towards cloud-based analytics with multiple users accessing shared datasets. This transition requires standardized, cleansed, and well-governed data to ensure scalability and security.

Why governance is essential for future initiatives:

- Enables trusted, analytics-ready datasets
- Supports controlled self-service analytics
- Reduces risk during cloud migration
- Improves collaboration across teams

2.6 Reducing Risk and Compliance Challenges

Manufacturing environments face increasing cybersecurity and compliance risks. A structured Data Governance and Management Framework help reduce these risks through clear accountability and control mechanisms.

Risk reduction through governance includes:

- Clear ownership and accountability for data
- Controlled access based on data criticality
- Auditability and monitoring of data usage
- Improved resilience against cyber threats

3. Data Governance & Management Framework for Michelin Sri Lanka

3.1 Goal of the Framework

The primary goal of this framework is to create a structured, secure, and scalable approach to managing Michelin Sri Lanka's data, ensuring high quality, strong protection of intellectual property, smooth operational performance, and readiness for advanced analytics.

It aligns data handling practices with the company's manufacturing, quality, and cloud-migration needs.

Key goals:

- Protect sensitive tyre recipe/formulation data
- Improve data accuracy, consistency, and traceability
- Strengthen accountability and reduce operational risks
- Enable efficient analytics and digitization efforts
- Standardize processes across the entire data lifecycle

3.2 Why These Domains Were Selected

Because Michelin Sri Lanka operates a complex manufacturing plant with high data volume and IP sensitivity, the selected domains prioritize **data protection**, **quality**, and **architecture stability**.

Priority reasons:

- Sensitive recipe data requires strict governance
- Quality and traceability depend on clean and reliable data
- Manufacturing analytics need consistent data models
- Interconnected systems demand a unified architecture
- Cloud migration needs controlled and secure data flows

3.3 Alignment with Standard Frameworks

The design is guided by globally accepted frameworks to maintain industry-grade quality and structure.

Referenced frameworks:

- DAMA-DMBOK (for domain structure + data lifecycle management)
- ISO 27001 (for data security controls)
- ISO 8000 (for data quality standards)
- NIST Cybersecurity Framework (for risk and protection alignment)

3.4 Overview of the Proposed Framework

This section presents the **customized Data Governance & Management Framework** for Michelin Sri Lanka, integrating governance, management, and analytics layers with practical policies, roles, and processes.

A. Data Governance Component

3.4.1 Data Governance Roles & Responsibilities

A clear role structure ensures accountability and controlled access—especially for recipe and quality data.

Role	Composition / Domain	Key Responsibilities
Governance Council	Plant Manager, IT Director, Production Director, Quality Director, R&D Manager	Approve policies, resolve conflicts, review performance, ensure alignment
Chief Data Steward	Program coordinator	Coordinate governance activities, facilitate policy development, monitor performance
Data Owner	Department heads	Define business rules and quality standards, approve access, resolve issues
Data Steward	Subject matter experts	Maintain dictionaries and metadata, monitor quality, investigate issues
Data Custodian	IT/OT teams	Implement controls, backups, validation rules, maintain system documentation
Data Consumer	All authorized employees and systems	Use data properly, report quality issues, follow governance procedures

3.4.2 Governance Policies

Policies guide how data is handled across its lifecycle.

Data Classification Policy:

- Classify all data assets based on criticality, sensitivity, and regulatory requirements
- **Classification Table**

Classification	Data Type	Impact	Retention
Super Critical	Rubber compound recipes, R&D formulations, product specifications, proprietary processes	Severe competitive disadvantage / IP loss	Permanent with version control
Critical	Production data, quality results, machine parameters, customer orders	Operational disruption, quality issues, compliance failures	7–10 years
Normal	Aggregated KPIs, reports, training materials, documentation	Minor impact	3–5 years or as needed

Access & Security Policy:

- Principle of least privilege
- Role-Based Access Control (RBAC) with quarterly reviews
- Dual authorization for super critical data
- Multi-factor authentication for Critical/Super Critical data
- Masking or anonymization for production/testing data

Data Quality Policy:

- Standards for accuracy, completeness, consistency, validity
- Sensor calibration with quarterly validation
- Serial numbers mandatory with automated validation
- Reference data centrally managed
- Real-time production data within 30 seconds
- Automated validation rules for anomalies

Naming Conventions:

Category	Standard Code	Example
Equipment	ML	ML_01_TEMP
Plant Location	LK	LK_PROD_001
Temperature	°C	CURE_TEMP=165°C
Time Duration	SEC, MIN	MIX_TIME=300 SEC
Quality Status	PASS/FAIL	VISUAL_INSPECT=PASS

Metadata Requirements:

- Business: name, definition, owner, steward, classification, intended use
- Technical: data type, format, units, storage location

- Operational: creation date, last modified, refresh frequency, quality metrics
- Lineage: source, transformations, dependencies, downstream users

Data Quality Thresholds:

Dimension	Measurement	Threshold
Completeness	% populated fields	≥99.5% Super Critical, ≥98% Critical
Accuracy	% within tolerance	≥99% sensors, 100% recipes
Consistency	% cross-system match	≥99% master data sync
Timeliness	Time from generation	≤30 sec real-time, ≤4 hrs batch

Compliance Monitoring & Audit

- Access, quality, and security controls monitored in real-time
- Periodic reviews: monthly, quarterly, semi-annual, annual
- Audit trails capture authentication, changes, incidents; retained by classification (Super Critical: 10 yrs, Critical: 7 yrs, Normal: 3 yrs)

RACI Model Example

Activity	Council	Steward	Owner	Custodian	Consumer
Policy Approval	A	R	C	C	I
Data Classification	I	C	A	R	I
Access Approval	I	C	A	C	R
Quality Monitoring	I	C	A	R	I

B. Data Management Component

3.4.4 Data Architecture

Layered architecture: Sensors → PLC → SCADA → MES → ERP → Data Lake

- Segregation: raw, processed, curated data
- Middleware for legacy integration
- Cloud-ready analytics layer
- ETL pipelines for validation
- Standardized schemas for products, machines, and quality datasets

3.4.5 Data Quality Management

Controls:

- Validation at SCADA/MES
- Completeness checks (serial numbers mandatory)
- Consistency checks across machines and factories
- Automated dashboards for monitoring
- KPIs: accuracy, completeness, timeliness, consistency

3.4.6 Master Data Management (MDM)

Key domains:

- Product Master: tyre codes, SKUs, specifications
- Recipe/Formulation Master: compound recipes, curing parameters
- Equipment Master: machine IDs, calibration
- Plant/Location Master: line IDs, warehouse zones
- Supplier Master: IDs, materials, contracts

Governance rules:

- Single Source of Truth
- Approval workflows for Critical data
- Version control for recipes and formulations
- Duplicate prevention
- Audit logging
- Periodic reviews

Lifecycle stages: Create → Validate → Approve → Publish → Maintain → Retire/Archive

Quality thresholds: Completeness ≥ 99%, Accuracy ≥ 99%, Consistency ≥ 99%, Uniqueness 100%, Timeliness ≤ 4 hrs.

3.4.7 Metadata Management

Aspect	Details
Metadata Types	Business, Technical, Operational, Security, Lineage
Cataloguing Requirements	Centralized data dictionary including: name, definition, owner, classification, source, storage, format, refresh frequency, retention, consumers
Lineage Tracking	Sensors → SCADA → MES → ERP → Data Lake → Power BI
Metadata Quality Monitoring	Completeness ≥ 98%, Definition Accuracy 100%, Lineage ≥ 95%, Naming Standard Compliance ≥ 99%

3.4.8 Data Security & Privacy

Encryption:

- AES-256 for data at rest
- TLS 1.2+ for transit
- Separate keys for Super Critical, Critical, backup data

Access Control:

RBAC with principle of least privilege

Dual authorization for Super Critical

Quarterly reviews and logging

Anonymization & Masking:

- Masking for analytics, training, audits, cloud
- Tokenization for serials
- Aggregation for KPIs
- Pseudonymization for supplier/customer identifiers

3.4.9 Data Lifecycle Management

Retention Policy:

Classification	Retention	Rationale
Super Critical	Permanent with version control	IP protection
Critical	7–10 yrs	Compliance, traceability
Normal	3–5 yrs	Operational reference

Archiving & Disposal:

- Encrypted, compressed, indexed archives
- Secure disposal via cryptographic erasure, multi-pass wiping
- Physical media destroyed securely with certificates
- Oversight via Governance Council

C. Analytics & Reporting Layer

- Governed Power BI datasets for plant/product performance
- Dashboards for root-cause analysis, management, and audit reports
- Future cloud-based self-service analytics with controlled access

3.4.10 Steps to Design & Implement

1. **Assessment:** map data flows, identify pain points
2. **Policy & Roles:** assign owners/stewards, define access/criticality
3. **Architecture Design:** standard schemas, middleware, cloud roadmap
4. **Quality & Security Controls:** validation rules, encryption, RBAC
5. **Monitoring & Continuous Improvement:** KPI-based audits, regular refinement

Tools & Standards: DAMA-DMBOK, COBIT, ISO 27001, Layered Data Lake, Power BI, ERP–MES–SCADA middleware

4. Phased Implementation Roadmap

4.1 Purpose of a Phased Approach

Implementing Data Governance and Management across a manufacturing organization like **Michelin Sri Lanka** is complex and cannot be done in a single step. A phased approach reduces risk, allows learning, and ensures minimal disruption to production operations.

Why a phased rollout is necessary:

- High operational dependency on data
- Presence of legacy systems
- Need to balance governance with production continuity
- Limited local support resources
- Gradual cultural adoption of governance practices

4.2 Phase 1 – Foundation & Data Governance Setup (0–6 Months)

This phase focuses on establishing the **core governance structure** and standardizing high-risk data elements before introducing technical complexity.

Key activities:

- Form a Data Governance Committee
- Identify and assign Data Owners and Data Stewards
- Classify data into super critical, critical, and normal
- Define enterprise-wide naming conventions and reference standards
- Document core data policies (access, quality, security)
- Identify high-priority data domains (recipes, quality, traceability)

Expected outcomes:

- Clear accountability and ownership
- Reduced ambiguity in data responsibility
- Initial standardization across systems

4.3 Phase 2 – Data Quality, Master Data, and Architecture Control (6–12 Months)

Once governance foundations are in place, the organization can focus on **improving data quality and consistency** across systems.

Key activities:

- Implement Master Data Management for products, recipes, and machines
- Standardize schemas across ERP, MES, and analytics platforms
- Introduce automated data quality validation rules
- Establish centralized metadata and data dictionary
- Improve integration with legacy systems using standardized middleware

Expected outcomes:

- Improved data consistency across machines and factories
- Reduced reliance on complex manual data dictionaries
- Higher trust in production and quality data

4.4 Phase 3 – Analytics Enablement & Cloud Readiness (12–18 Months)

This phase focuses on unlocking **business value from governed data** while preparing the organization for scalable analytics and future digital initiatives.

Key activities:

- Create curated, analytics-ready datasets
- Certify KPIs for plant and product performance
- Enable governed self-service analytics using Power BI
- Implement controlled cloud data storage for cleansed data
- Apply anonymization techniques for broader analytics access

Expected outcomes:

- Single source of truth for performance metrics
- Faster and more accurate decision-making
- Secure and scalable analytics environment

4.5 Phase 4 – Optimization & Continuous Improvement (18+ Months)

Data Governance is an ongoing process that evolves with business and technology changes.

Key activities:

- Regular data quality and access audits
- Continuous refinement of governance policies
- Onboarding new machines and factories into standards
- Advanced analytics and predictive maintenance use cases
- Ongoing training and awareness programs

Expected outcomes:

- Mature data governance culture
- Continuous reduction in data-related risks
- Sustained business value from data assets

4.6 Success Measurement and Governance KPIs

To ensure effective implementation, governance performance must be measured.

Sample governance KPIs:

- Data quality score improvement
- Reduction in data-related production issues
- Time taken for root-cause analysis
- Compliance with data standards
- User adoption of governed datasets

5. Business Benefits of the Proposed Framework

5.1 Improved Product Quality and Traceability

By standardizing data and implementing strict governance on production parameters and recipes, Michelin can ensure high product quality and full traceability.

Benefits:

- Faster identification of defects or process issues
- Accurate root-cause analysis for complaints
- Reduced risk of production errors and recalls
- Enhanced customer trust and satisfaction

5.2 Enhanced Operational Efficiency

Governed and standardized data reduces manual corrections and inconsistencies, improving day-to-day plant operations.

Benefits:

- Streamlined production planning and scheduling
- Consistent KPI measurement across plants
- Reduced downtime due to data errors
- Optimized resource allocation for raw materials and labour

5.3 Protection of Intellectual Property

Critical recipe and formulation data is classified and secured, protecting Michelin's competitive advantage.

Benefits:

- Controlled access to sensitive recipe information
- Audit trails for recipe changes ensure accountability
- Reduced risk of data leaks or unauthorized modifications
- Secure foundation for future R&D initiatives

5.4 Better Decision-Making and Analytics Readiness

Curated, quality-controlled data enables accurate analytics and informed management decisions.

Benefits:

- Reliable performance dashboards for plant and product metrics
- Faster, evidence-based decision-making
- Supports predictive analytics and future cloud-based initiatives
- Facilitates cross-department collaboration with a single source of truth

5.5 Regulatory Compliance and Risk Reduction

Structured governance helps manage compliance, security, and operational risks.

Benefits:

- Easier audits and compliance with internal and industry standards
- Reduced cybersecurity risks through access controls and encryption
- Minimizes data inconsistencies and operational disruptions
- Supports scalable, secure expansion and modernization

6. Conclusion

Michelin Sri Lanka operates in a highly complex and data-intensive manufacturing environment. The proposed Data Governance and Management Framework ensures that critical data, such as product recipes, machine parameters, and quality metrics, is accurate, secure, and traceable throughout its lifecycle.

By implementing structured governance roles, standardized processes, and layered data management practices, the organization can reduce operational risks, enhance product quality, protect intellectual property, and enable reliable analytics.

- Governance provides clear accountability and reduces ambiguity in data handling
- Standardization and master data management ensure consistency across factories and systems
- Data quality, security, and lifecycle controls enable safer, more reliable operations
- Phased implementation ensures practical adoption and sustainable improvements
- The framework supports both current operations and future digital initiatives such as cloud analytics

In summary, this framework equips Michelin Sri Lanka to leverage its data as a strategic asset, improving decision-making, operational efficiency, and competitiveness while maintaining high standards of quality and security.